# Holonomic D-modules, duality, functors

**Def:** $D(X)$ = derived category of $\mathcal{D}_X$-modules on $X$ w/ quasicoh. cohomology.
$\simeq$ derived cat of quasicoh $\mathcal{D}_X$-mod on $X$

cohomological grading: diff. inc. degree

$D^+(X)$ = bounded below category (injective resolutions)
cohomology $0 \to M \to I^0 \to I^1 \to I^2 \to \cdots$

$D^-(X)$ = bounded above cat

$D^b_c(X)$ = cohomology bounded, $\mathcal{D}_X$-cohomology

Question: How does S.S transform under functors?

singular supp: choose a "good" filtration $F_\bullet M$

$gr_\bullet M \in QCoh(T^*_X)$

$SS(M) = supp(gr(M)) \subseteq T^*_X$

**Prop** Let $i: Z \overset{cl}{\hookrightarrow} X$

$$SS(M) \longmapsto SS(i_\bullet M)$$

$$\begin{array}{ccccc} \cap\!\!\shortmid & & & & \\ T^*_Z & \longleftarrow & T^*_X \times_X Z & \hookrightarrow & T^*_X \end{array}$$

**Proof:** local calculation, $z \in Z$

on local rings. $i_\bullet M = \mathcal{D}_X/\mathcal{D}_X I \otimes_{\mathcal{D}_Z} M$ ← good filtration: $\otimes$ of • $\mathcal{D}_X/\mathcal{D}_X I$, order filtration • $M$ good filtration.

Claim: $gr(M \otimes_A N) = gr(M) \otimes_{gr(A)} gr(N)$ [flat over $A_0$?]

$F_n(M \otimes_A N) = \sum_{i+j=n} (F_i M \otimes_{A_0} F_j M)$

$gr(\mathcal{D}_X/\mathcal{D}_X I) = \mathcal{O}_{T^*_X}|_Z = \mathcal{O}_{T^*_X \times_X Z}$

$supp(gr(i_\bullet M)) = (T^*_X \times_X Z) \times_{T^*_Z} SS(M)$

$M$ gets acted on by $x_1, \dots, x_r$, $\partial_1, \dots, \partial_r$

Know: $I = (x_{r+1}, \dots, x_n)$ act by torsion on $M$.

$supp(M) \subseteq Z$

Check: $\partial_1, \dots, \partial_n$ → $\partial_1, \dots, \partial_r$ will act through $M$

→ $\partial_{r+1}, \dots, \partial_n$ act nontrivial

$\xi_{r+1}, \dots, \xi_n \notin Ann$.

**Prop:** $p: X \to Y$ smooth, $M \in DMod(Y)$

$$\begin{array}{ccccc} SS(M) & \longrightarrow & & & SS(f^+M) \\ \cap\!\!\shortmid & & & & \cap\!\!\shortmid \\ T^*_Y & \longleftarrow & T^*_Y \times_Y X & \overset{d}{\hookrightarrow} & T^*_X \end{array}$$

defined by vertical tangents

**Proof:** $p^+M = p^*M$ $\mathcal{O}$-pullback.

trivial filtration $F_i := \mathcal{O}_X \otimes F_i M$ $= \mathcal{O}_X \otimes_{f^*\mathcal{O}_Y} f^*M$

Claim: vertical tangents act by torsion

$\theta \cdot (f \otimes m) = \underbrace{\theta \cdot f}_{\text{order } 1} \otimes m + \underbrace{f \otimes \tilde{\theta} \cdot m}_{\text{torsion}}$ $= \theta \cdot f \otimes m$ $= 0$ in $gr$

$\tilde{\theta}$ push of tangents to $Y$

If Hein it: $\mathrm{supp}(\mathrm{gr}(p^{\dagger}\mathcal{M})) = X \times_Y SS(\mathcal{M})$

$$\begin{tikzcd} \square \arrow[d] & Y \\ X \arrow[r] & \end{tikzcd}$$

**Thm** $\mathcal{M} \in D\mathrm{Mod}(X)$

$$\boxed{\dim(SS(\mathcal{M})) \geq \dim(X) = \frac{1}{2}\dim(T^*X)}$$

**Pf:** suppose that $\mathrm{supp}(\mathcal{M}) = r < n$

not smooth.

then, $\mathcal{M}$ is supported on $Z \underset{cl.}{\subseteq} X$ $\quad \dim(Z) = r$

Kashiwara: $\mathcal{M} = i_\circ \mathcal{M}_0$ where $\mathcal{M}_0 \in D\mathrm{Mod}(Z)$

If $Z$ not smooth choose open $U \subseteq X$ where $U \cap Z$ is smooth.

Assume $Z$ smooth; by above, $\dim(SS(i_\circ \mathcal{M}_0)) = \dim(SS(\mathcal{M}_0)) + \mathrm{codim}(Z)$

$= \dim(SS(\mathcal{M}_0)) + (n-r)$

$n = \dim(X)$

By induction, $\dim(SS(\mathcal{M}_0)) \geq r$

So, $\dim(i_\circ \mathcal{M}_0) \geq r + n - r = n$. $\square$

**Def:** $\mathcal{M}$ is h. holonomic if

$\dim(SS(\mathcal{M})) = \dim(X)$. i.e. minimum possible.

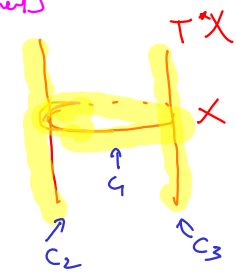
**Def'n** $\mathcal{M}$ D-module, $SS(\mathcal{M})^{red} = \bigcup C_i$ irreducible components

Let $\eta_i$ be generic point of $C_i$

good $\longrightarrow$ $\mathrm{gr}(\mathcal{M}, F)_{\eta_i}$ module for $(\mathcal{O}_{T^*X})_{\eta_i}$

By Nakayama, it's finite length. (reduce mod $F_i$)

$SS(\mathcal{M}) =$




**Defn:** $m_i = \mathrm{length}(\mathrm{gr}(\mathcal{M}, F)_{\eta})$

$CC(\mathcal{M}) = \sum m_i \cdot C_i$

**Claim:** - independent of good filtration.

- $\mathrm{supp}(CC(\mathcal{M})) = SS(\mathcal{M})$

**Ex.** $0 \to D/D\partial \xrightarrow{\cdot\partial} D/D\partial^2 \xrightarrow{\cdot\partial} D/D\partial \to 0$

$1 \mapsto \partial$

$1 \mapsto 1$

$X = \mathbb{A}^1$

$SS(D/\partial^2)$ $\qquad D/\partial^2$

$0 \subseteq \mathcal{O}_X \cdot 1 \subseteq \mathcal{O}_X \cdot 1 \oplus \mathcal{O}_X \cdot \partial \subseteq \cdots$

$\partial \cdot f \qquad f\partial r$

$SS(D/\partial^2) = X = SS(D/\partial)$

$CC(D/\partial^2) = 2 \cdot X \neq SS(D/\partial)$

$\mathcal{O}_X \oplus \mathcal{O}_X \oplus 0 \cong \dfrac{\mathcal{O}_X[\xi]}{\xi^2}$

$\xi \qquad \xi$

multiplicity 2.

**Prop:** • CC is additive w.r. S.E.S.

• holonomic D-modules have finite length

• generally flat conn

(Bernstein dim): $N$ acts by finitely many vectors on $U/B$

$\Rightarrow \Lambda = \bigcup_{\text{finite}} N^{\alpha}$ vectors $U/B$ module components dim $n$ $\rightsquigarrow$ $DMod_{\Lambda}(U/B)$

$DMod(N \curvearrowright U/B)$

$\|$ are all holonomic

**Ex:** all D-modules from SL2 exercises are holonomic

- $\mathcal{O}_X$ is holonomic
- $i_! \mathcal{O}_Z$ is holonomic.

NON-ex: $D_X$

**Remark:** flat conn $\underset{\text{def}}{\Longleftrightarrow}$ $\mathcal{O}_X$-locally free D-module $\overset{\text{Cor}}{\Longleftrightarrow}$ $\mathcal{O}_X$-coh D-mod.

i.e. local vb w/ connection.

$\uparrow$ D-mod.

$\mathbb{P}^1$

$\circlearrowleft \alpha$

$0 \to \delta_{\alpha} \to \cdots ? \to \alpha_* \delta_{\alpha(\alpha)} \to ? \to 0$

$0 \to M_1 \to M_2 \to M_3 \to 0$

$CC(M_2) = CC(M_1) + CC(M_3)$